

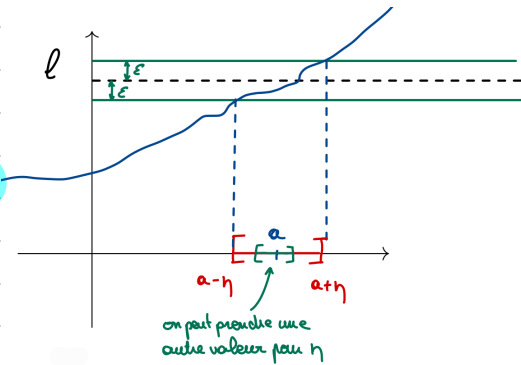
LIMITE FINIE

Soit $f: I \rightarrow \mathbb{R}$ et $l \in \mathbb{R}$

On dit que f admet l comme limite en a si :

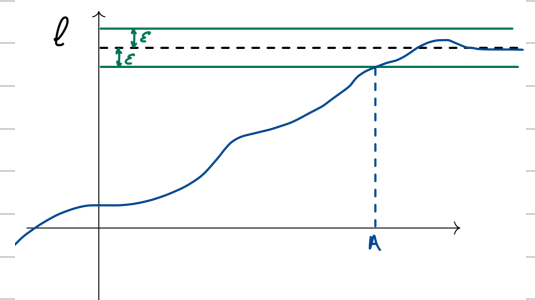
- Dans le cas $a \in \mathbb{R}$:

$$\forall \varepsilon > 0, \exists \eta > 0, \forall x \in I, |x - a| \leq \eta \Rightarrow |f(x) - l| \leq \varepsilon$$



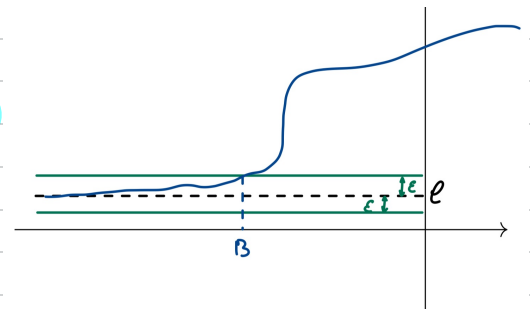
- Dans le cas où $a = +\infty$:

$$\forall \varepsilon > 0, \exists A \in \mathbb{R}, \forall x \in I, x \geq A \Rightarrow |f(x) - l| \leq \varepsilon$$



- Dans le cas où $a = -\infty$:

$$\forall \varepsilon > 0, \exists B \in \mathbb{R}, \forall x \in I, x \leq B \Rightarrow |f(x) - l| \leq \varepsilon$$



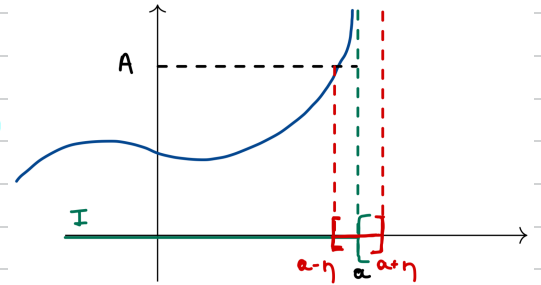
LIMITES INFINIES

Soit $f: I \rightarrow \mathbb{R}$.

On dit que f admet $+\infty$ comme limite en a si :

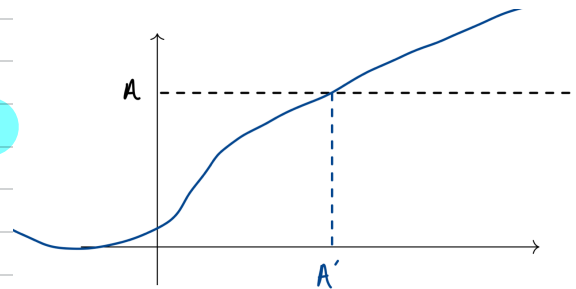
• Dans le cas $a \in \mathbb{R}$:

$$\forall A \in \mathbb{R}, \exists \eta > 0, \forall x \in I, |x - a| \leq \eta \Rightarrow f(x) \geq A$$



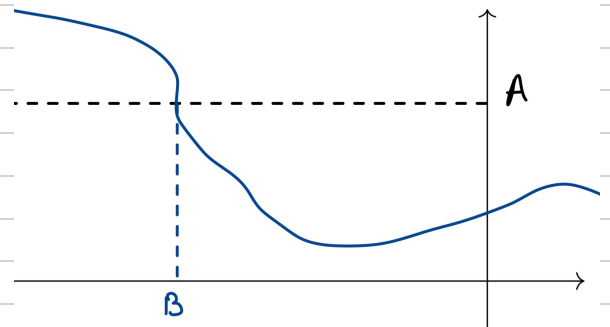
• Dans le cas $a = +\infty$:

$$\forall A \in \mathbb{R}, \exists A' \in \mathbb{R}, \forall x \in I, x \geq A' \Rightarrow f(x) \geq A$$



• Dans le cas $a = -\infty$:

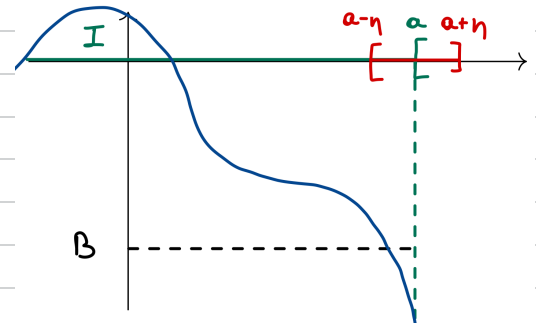
$$\forall A \in \mathbb{R}, \exists B \in \mathbb{R}, \forall x \in I, x \leq B \Rightarrow f(x) \geq A$$



On dit que f admet $-\infty$ comme limite en a si :

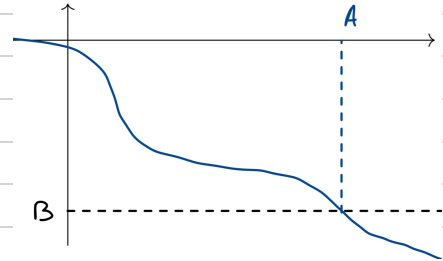
Dans le cas $a \in \mathbb{R}$:

$$\forall B \in \mathbb{R}, \exists \eta > 0, \forall x \in I, |x - a| \leq \eta \Rightarrow f(x) \leq B$$



Dans le cas $a = +\infty$:

$$\forall B \in \mathbb{R}, \exists A \in \mathbb{R}, \forall x \in I, x \geq A \Rightarrow f(x) \leq B$$



Dans le cas $a = -\infty$:

$$\forall B \in \mathbb{R}, \exists B' \in \mathbb{R}, \forall x \in I, x \leq B' \Rightarrow f(x) \leq B$$

